b)(40 p.) If the message signal is $m(t) = -5 + 10\sum_{n=-\infty}^{\infty} \Pi\left(\frac{t-n.4.10^{-3}}{2.10^{-3}}\right)$ and $f_c = 10\,kHz$, $k_f = 1500$,

b1) sketch the FM signal $y(t)$,

b2) find the power of FM signal,

b3) find the bandwidth (BW) of FM signal,

b4) if $m(t/2)$ is transmitted as message signal, find the BW and comment on it,

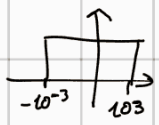
b5) find the maximum phase deviation $\Delta\emptyset$,

b6) sketch the demodulator and sketch the signal at each step.

(Hint: The Fourier series expansion of bipolar square wave (alternating between 1 V ve -1 V)= $\frac{4}{\pi}[Cos(\omega_0 t) - \frac{1}{3}Cos(3\omega_0 t) + \frac{1}{5}Cos(5\omega_0 t) - \frac{1}{7}Cos(7\omega_0 t) + \cdots])$

$$y_{FM}(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(z)\,dz\right)$$

$$\sum_{n=-\infty}^{\infty} \Pi\left(\frac{t - n(4.10^{-3})}{2.10^{-3}}\right) \rightarrow T_0 \qquad \Pi\left(\frac{t}{2.10^{-3}}\right) \rightarrow$$



$$T_0 = 4.10^{-3}\,sn$$

$$f_0 = \frac{1}{4.10^{-3}} = \frac{1000}{4}$$

$$= 250 Hz$$